nected to temperature sensors to halt the system in case the temperature rises above a threshold. If you have ever heard of phones shutting themselves down because the temperature was too high, or witnessed your phone shutting down because of low-battery, these gentle-chips were doing their job!

Graphics Processors (GPUs)

These are not found on all types of SoCs but only on the ones which need to perform graphic-intensive work. Smartphones are one of the most common SoC implementations which contain Graphics Processors. Graphics processors are not used on every system which needs to display something. A system with a simple LED display would not need one. Also, not all systems outputting data on bitmapped displays need a graphics processor. For example, a simple weather reader displaying weather and time on a bitmapped display does not need a dedicated GPU.

Graphics Processing units are useful on SoCs which need to perform tasks such as playing games, processing shadows, etc. If the system does not process 3D data, a dedicated GPU is not always required; they are power-hungry and can generate a lot of heat.

On-Chip Debug and Self-Test systems

Unlike software products, you can't just 'rewrite the code and push an update'. If any chip has a hardware-level bug, the company would have to fix the bug and resend the updated package. If the package was used to build a consumer gadget then it's even worse.

In the early days, debugging a chip meant that the tests were run using bond-out processor which had to be physically placed where the processor was supposed to be. Modern SoCs contain an On-Chip debug circuits which allow the developer to put the chip in debug mode where the behavior of processor changes and instead of depending directly on electronic clock-ticks, they start receiving ticks via the debugger unit which allows the programmer to inspect the variables and system state closely.

Summary

While we are running out of space here, we believe we have introduced the most generic hardware you can find on an SoC. But that's just the tip of the iceberg. If you were to start getting deeper into the terms introduced in this chapter, you would find yourself in an ocean of information. That said, we are going to move to the next chapter – about software on SoCs. d